StudDFT

Scientific-Educational Quantum Chemistry Program
(short manual, v1.30.2)

Contents

Introduction

StudDFT is a quantum chemical program developed for calculating small and medium-sized molecular systems using the Hartree–Fock (HF), density functional theory (DFT), and second-order Møller–Plesset perturbation theory (MP2) methods. The program performs energy calculations, geometry optimization with analytical gradients, Hessian calculations, vibrational frequencies and IR intensities, electronic and thermodynamic characteristics, transition state searches, and PES scanning, CDFT, TDDFT, RT-TDDFT, and Ehernfest MD. Its main purpose is to conduct educational quantum chemical calculations for undergraduate, graduate and postgraduate students studying quantum chemistry.

Author, program site, and citation

The program has been developed by Prof. Dr. Stanislav K. Ignatov, Professor of the Chemistry Department, Lobachevsky State University of Nizhny Novgorod, Russia, 2026.

The binaries of StudDFT for Windows and Linux systems can be download from https://github.com/skignatov/StudDFT-release

Please send your comments, questions, and reports of possible errors to skignatov@gmail.com

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Quick start

Jump to Section "*Obtaining program, setting up and start calculations*" to get started quickly

Editions (v1.14.0+)

Since v1.14.0 StudDFT ships in two editions built from the same source tree. The **FULL EDITION** contains every feature. The **STUDENT EDITION** excludes three advanced subsystems -- constrained

DFT (%cdft family), real-time TDDFT (%rt_* family) and Ehrenfest dynamics (%eh_* family); every other feature, keyword and numerical result is identical between the editions.

If a Student Edition encounters a Full-only keyword it stops with: "*** ERROR: %cdft is not available in the STUDENT edition of StudDFT; use the full build (make EDITION=full, or the full VS configuration)." The keyword reference below is split accordingly: Part I lists the keywords available in both editions, Part II the Full-Edition-only keywords.

Program capabilities (v1.17.1)

- **Electronic-structure methods.** Restricted and unrestricted HF; MP2/UMP2 with spin-projected PUMP2 (energies, analytical gradients, FD-of-gradient Hessians); DFT: LDA (SVWN, SVWN3), GGA (BLYP, BP86, BPW91, PBE), global hybrids (B3LYP/B3LYP3, PBE0), meta-GGA (r2SCAN), and range-separated hybrids (wB97, wB97X, wB97X-D with built-in CHG dispersion, tunable via %rs_omega). Open-shell systems automatically switch to the U-prefixed variants.
- **Job types.** Single-point energies; analytical gradients; geometry optimization (redundant-internals RFO or Cartesian BFGS) with frozen bond/angle/dihedral constraints; harmonic frequencies with analytical Hessians (HF/DFT, incl. closed-shell range-separated hybrids and pure-functional RI-J) or semi-numerical gradient-FD Hessians, IR intensities on the analytical route; opt+freq; Boys-Bernardi counterpoise (BSSE); rigid and relaxed PES scans over Z-matrix coordinates; transition-state search (P-RFO) with mode following and .hess restart; TS+freq; intrinsic reaction coordinate (IRC) following via the Gonzalez-Schlegel GS2 method; Born-Oppenheimer molecular dynamics (%job md, velocity-Verlet NVE) (v1.11.0).
- **Excited states (TDDFT/TDHF).** Linear-response TDDFT/TDHF (Casida) for closed-shell HF/DFT references; full RPA or Tamm-Dancoff; singlet/triplet/both; oscillator strengths; transition and unrelaxed state dipole moments with a Generalized Mulliken-Hush coupling table (%tddft_dipmom); MOLDEN + CI-amplitude export for MultiWfn/THEODORE (%export_tddft); LR-PCM solvent response with nonequilibrium or equilibrium solvation. Heavily accelerated in v1.15.0-1.16.0: the singlet LDA/GGA/hybrid kernel uses a rank-1 MO-amplitude fast path with a per-block XC cache, and all AO->MO half-transforms are dgemm-major Level-3 BLAS -- 30-80x faster end-to-end on transition-metal systems, identical spectra.
- **Real-time TDDFT (Full Edition, v1.10.0-1.11.x).** Explicit electron propagation for RKS/UKS and smeared references: exponential-midpoint PC, Magnus-2, ETRS, MMUT and EM propagators; delta-kick absorption spectra with Lorentzian-damped Fourier analysis; static or harmonic external fields with ramps and cutoffs (resonant-Rabi setups); moving external point charges (projectile protocols, v1.11.14); Mulliken and CDFT-fragment population trajectories with a per-atom charge-mode decomposition (density normal modes); the CDFT-release protocol (constrained preparation -> release -> free propagation) for charge-transfer dynamics.
- **Ehrenfest dynamics (Full Edition, v1.11.0-1.12.0).** Mean-field nonadiabatic MD with electronic substeps, NVE or thermally sampled initial conditions, comoving-basis treatment, force-check oracle modes; v1.12.0 adds the smeared Ehrenfest scheme: exact spectral transport of fractional occupations for metallic/smeared references (RKS and UKS).
- **Implicit solvation (PCM).** IEFPCM (default), C-PCM, and COSMO with named-solvent table or explicit permittivity; single points, analytical PCM gradients, optimizations, and semi-numerical PCM frequencies for HF and all DFT classes (RKS and UKS); LR-PCM coupling for TDDFT.

- **Empirical dispersion.** Native clean-room D3(BJ) implementation (energies, gradients, Hessians) for HF and common functionals (v1.13.0) and a native Chai-Head-Gordon CHG tail for wB97X-D covering H-Xe (v1.13.1-1.13.2); all external LGPL dispersion components were removed in the v1.12.1 licensing clean-up (D4 is excluded from current builds).
- **Constrained DFT (Full Edition).** Wu-Van Voorhis charge-constrained DFT on Becke atomic fragments (single-point energies; restricted or unrestricted reference; multiple simultaneous constraints; two-loop Newton solver with analytic response Jacobian); checkpoint warm starts (%cdft_initial_guess); 1e-integral and MO export tailored for diabatic coupling (S_AB, H_AB) construction; the RT CDFT-release protocol.
- **Metallic and difficult SCF.** Fractional-occupation (Mermin) smearing -- Fermi-Dirac, Gaussian, Methfessel-Paxton (N=1), and Marzari-Vanderbilt cold smearing -- with fixed- or relaxed-moment UKS conventions; analytical smeared gradients, geometry optimization and semi-numerical frequencies on the Mermin free-energy surface; self-tuning damping + DIIS stabiliser for smeared metallic SCFs; AUTO level shift; SAD, core-Hamiltonian, or checkpoint initial guesses; MOM/Aufbau occupation control; SCF checkpoint / warm-restart machinery (%chkfile, %chk_write, %chk_every, %initial_guess read, %geo read).
- **Performance.** RI-J and RI-JK density fitting (def2-universal JFIT/JKFIT auxiliary sets; JFIT extended to K-Rn) with analytical RI gradients and an analytic RI-J Hessian for pure closed-shell functionals; direct and incremental (delta-P) SCF with an ERI-store memory cap and allocation guards (v1.13.3); per-quartet auto HRR dispatch in the ERI kernels; optional multigrid SCF; OpenMP parallelism across the whole hot path (1e integrals, JK, XC quadrature, gradients, CPHF/CPKS, TDDFT); symmetry-blocked numerical Hessians displace only symmetry-unique atoms.
- **ECPs.** Effective core potentials (built-in LANL2DZ, per-atom MIXED assignment) with numerical (Mura-Knowles quadrature) or analytical (McMurchie-Davidson) integral evaluation, including gradients and Hessians.
- **Analysis and output.** Mulliken charges plus extended population analysis (Lowdin, Hirshfeld, CM5 charges; Mayer/Mulliken/Lowdin bond orders; Mayer valences; spin populations); always-on point-group detection with irreducible-representation labels for MOs and vibrational modes behind %symmetry; static dipole polarizability from field CPKS; finite external electric field; MOLDEN orbital and normal-mode export; .hess files; 1e-integral print/export; per-timer profiling.
- **Basis sets.** Cartesian (default) or spherical-harmonic shells; built-in Pople and Dunning sets from STO-3G to 6-311++G(3df,2pd) and (aug-)cc-pVDZ/TZ/QZ plus LANL2DZ; per-atom MIXED basis assignment.

Keyword reference

In the tables below, alternatives are separated by "|" and the default option is shown in square brackets or marked "default". Keywords, options and synonyms are case-insensitive (file names preserve case). Version tags (vX.YY.Z) mark the release in which a feature or its current behaviour was introduced.

Part I. Keywords available in both editions (Student and Full)

Quantum chemical theories

Keyword	Options	Synonyms	Designation
%method			Required. Electronic-structure method (case-insensitive).
	HF		Hartree–Fock
	MP2		HF reference + MP2 correlation (post-SCF); spin-projected PUMP2 for open shells (see %pump2)
	SVWN	SVWN5, LDA, LSDA	Slater + VWN-V LDA (NWChem / PySCF convention)
	SVWN3		Slater + VWN-III RPA (Gaussian 09 convention)
	BLYP		Becke88 + LYP GGA
	BP86	BP	Becke88 + Perdew86 GGA
	BPW91		Becke88 + Perdew–Wang 91 GGA
	PBE		Perdew–Burke–Ernzerhof GGA
	PBE0	PBE1PBE	PBE with 25 % HF exchange
	B3LYP	B3LYP5	B3LYP with VWN-V correlation (ORCA / NWChem)
	B3LYP3		B3LYP with VWN-III correlation (Gaussian convention)
	R2SCAN		r2SCAN meta-GGA (v0.79+). Depends on rho, |grad rho|^2 and tau.
	wB97		Range-separated hybrid, 100 % long-range HF exchange (alpha=0, beta=1, omega=0.4) (v0.88+)
	wB97X		Range-separated hybrid with 15.77 % short-range HF exchange (omega=0.3) (v0.88+)
	wB97X-D	wB97XD	wB97X re-fit with built-in Chai–Head-Gordon damped C6/R^6 dispersion (omega=0.2); the CHG tail is enabled automatically (v0.88.1+)
			Note: open-shell calculations automatically use U-prefixed variants (UHF, UMP2, UB3LYP, ...). RS functionals support energy / gradient / opt / scan / ts / irc / numerical freq; analytic RS Hessian and TDDFT are closed-shell (RKS) only.
%rs_omega	<positive real>		Override of the range-separation parameter omega (bohr^-1) for the wB97 family ("omega-tuning"; retunes both the exchange splitting and the SR-DFT attenuation) (v0.88.0). Error if the method is not range-separated.

Job type

Keyword	Options	Synonyms	Designation
%job			Task to run. Default: energy.
	energy	sp, single	Single-point energy (default)
	gradient	grad, force	Analytical gradient at fixed geometry
	opt	optimize	Geometry optimization (internals RFO or Cartesian BFGS, see %opt_coord)
	freq	frequency, hessian	Hessian + harmonic vibrational frequencies; writes <base>.hess and <base>.molden
	opt+freq	optfreq	Opt followed by freq at the optimized geometry
	bsse	counterpoise, cp	Boys–Bernardi counterpoise (requires fragment IDs as a 5th column on every geometry line; fragments must be numbered contiguously from 1)
	scan	pescan, rigid_scan	Rigid PES scan; requires %gz block and %scan ... *end block
	scanopt	relaxed_scan, scan+opt	Relaxed PES scan: at every grid point a constrained geometry optimization is performed
	ts	opt_ts, optts	Transition-state search via P-RFO (HF / DFT only)
	ts+freq	tsfreq, opt_ts+freq, opttsfreq	TS search followed by freq pass at located TS; writes <base>_ts.hess / <base>_ts.molden
	irc		Intrinsic-reaction-coordinate walk (Gonzalez–Schlegel GS2) from a stored TS Hessian; requires %read_hess (v0.86.0). Gas-phase HF / DFT only (no MP2 / PCM / CDFT). Writes <base>_irc_forward.xyz / <base>_irc_reverse.xyz trajectories.
	md	bomd	Born-Oppenheimer molecular dynamics on the ground-state surface (velocity-Verlet NVE) (v1.11.0+)

Molecule builder

Keyword	Options	Synonyms	Designation
%charge	<signed integer>		Total molecular charge. Default: 0.
%mult	<positive integer>	%multiplicity	Spin multiplicity 2S+1 (1 = singlet, 2 = doublet, ...). Default: 1.

Keyword	Options	Synonyms	Designation
%spin	restricted | unrestricted	RHF, RKS, R | UHF, UKS, U	SCF spin treatment. If not set, determined automatically from %mult (open shells switch to unrestricted).
%geo	[angstrom | bohr] | read	%geometry, *geometry | read: restart	Start of Cartesian geometry block. Terminated by *end or empty line. Atom token: <Z> | <Element> | <Element><index>. "%geo read" (v1.08.2) takes the element list and coordinates from the checkpoint file (see %chkfile) instead of the block; any atom lines still present are parsed and discarded with a NOTE. Since v1.30.2 the same block also accepts a Z-MATRIX: when its first data line carries exactly ONE token the block is read as a Z-matrix and %gz is not needed. The two formats cannot collide - a Cartesian line always has 4 tokens (5 under %job bsse) while Z-matrix rows 1 / 2 / 3 / 4+ have 1 / 3 / 5 / 7 - and a one-token line was a parse error before, so no working input changes meaning. In-line comments are stripped before the count is taken. An auto-detected Z-matrix closes on *end as well as on a blank line. "%geo bohr" then draws a NOTE that it does not apply (Z-matrix lengths are always Angstroems, angles always degrees) and is ignored.
%gz			Start of Z-matrix geometry block (Angstroems, degrees). Terminated by empty line. Since v1.30.2 the keyword itself is OPTIONAL: a Z-matrix written inside an ordinary %geo block is detected automatically (see %geo), and such a block closes on *end as well as on a blank line. %job scan / scanopt still require a Z-matrix geometry - from either spelling - because they sweep Z-matrix coordinates.

Basis set

Keyword	Options	Synonyms	Designation
%basis	<basis set name>		Required. Basis set name (case-insensitive). Built-in: sto-3g, 3-21g, 6-31g, 6-31g(d), 6-31g(d,p), 6-31+g(d), 6-31++g(d,p), 6-311g(d), 6-311g(d,p), 6-311g(2d,2p), 6-311g(3df,2pd), 6-311++g(2d,2p), 6-311++g(3df,2pd), cc-pVDZ, cc-pVTZ, cc-pVQZ, aug-cc-pVDZ, aug-cc-pVTZ, aug-cc-pVQZ, LANL2DZ. Pople star forms are resolved automatically (6-31G* = 6-31G(d), 6-31G** = 6-31G(d,p)). MIXED: per-atom

Keyword	Options	Synonyms	Designation
			assignment via <selector> <basis_name> lines until next %-keyword or empty line.
%spherical	(flag)		Use spherical-harmonic d/f-shells (5d/7f); default is Cartesian (6d/10f).

ECP

Keyword	Options	Synonyms	Designation
%ecp	<ecp name>		Effective core potential. Built-in: LANL2DZ. MIXED: per-atom assignment via <selector> <ecp_name> lines until next %-keyword or empty line.
%ecp_method	[numerical] | analytical	numerical: quadrature, num, 0 | analytical: md, mcmurchie, 1	ECP integral evaluation method. Default: numerical (Mura–Knowles radial quadrature). Analytical = McMurchie–Davidson (matches g09 to ~1 microHa on AlH3-class molecules since v0.77.1). For very tight basis primitives (alpha > ~5000) prefer numerical.

Electronic integral calculations

Keyword	Options	Synonyms	Designation
%boys_cook	(flag)	%boys_precise	Use Cook recurrence for Boys function instead of fast tabulated representation. Default: off (table method).
%hrr_mode	[auto] | contracted | primitive	contracted: new, fast | primitive: old, legacy	HRR placement in the HGP ERI kernels (v0.91–0.95). AUTO (default) picks the cheaper kernel per shell quartet (contraction-first iff the quartet contraction product Kq >= 2). "contracted" / "primitive" force one path; primitive reproduces historical references bit-exactly.

SCF procedure

Keyword	Options	Synonyms	Designation
%scf_converger	[DIIS] | NONE	%converger	Convergence accelerator. Default: DIIS. Internally tests whether the value starts with "DIIS"; anything else turns DIIS off.
%scf_maxiter	<positive integer>		Max number of SCF iterations. Default: 150. Also propagated to the SAD atomic SCFs (effective

Keyword	Options	Synonyms	Designation
			atomic budget max(50, %scf_maxiter), v1.07.2).
%scf_etol	<positive real>		Energy convergence threshold (Hartree). Default: 1e-8.
%scf_ptol	<positive real>		Density-matrix convergence threshold. Default: 1e-6.
%scf_diis_etol	<positive real>	%scf_gtol	DIIS error-norm (orbital-gradient) convergence threshold (v0.76.0). All three (%scf_etol, %scf_ptol, this one) must be satisfied simultaneously. Default: 1e-5.
%scf_min_iter	<positive integer>		Minimum SCF iterations before convergence can be declared (v0.76.0). Default: 5.
%scf_n_converged	<positive integer>	%scf_nconv	Consecutive iterations all three convergence criteria must be satisfied (v0.76.0). Default: 2.
%scf_diis_cond_max	<real > 1>	%scf_diis_condmax	Upper bound on DIIS B-matrix condition number; oldest vector is pruned when exceeded (v0.76.0). Default: 1e10.
%damping	<real in [0,1]>		Density-matrix damping factor. DIIS overrides damping after a few iterations. Default: 0.0. (Smeared metallic SCFs use their own self-tuning damping, v1.07.4.)
%level_shift	AUTO | OFF | <non-negative real>		Virtual-orbital level shift (Hartree). Useful for problematic open-shell / heavy-element cases. Default: AUTO (turning on shift 0.2 if SCF is not converged in 100 steps). Forced OFF under %smearing.
%initial_guess	AUTO | SAD | CORE | READ	%guess, %scf_guess; HCORE, CORE_H; READ: chk, restart	Initial-guess strategy (v0.76.0). Default: AUTO (RHF/RKS -> CORE; UHF/UKS -> SAD if a fit basis is loadable, CORE otherwise). SAD supports ECP atoms. READ (v1.08.0) seeds the SCF from the checkpoint file (%chkfile); a failed read is fatal by design. Cross-spin/-

Keyword	Options	Synonyms	Designation
			multiplicity reads convert correctly (warm restarts for sigma-ladders and multiplicity scans).
%uks_occupation	AUTO | STRICT | MOM | MIX	%uhf_occupation, %occupation; AUFBAU, MAXOV, MAX_OVERLAP, REMIX	UHF/UKS orbital-occupation strategy (v0.76.0). Default: AUTO (MOM for the first iterations, then strict Aufbau). No effect for restricted; bypassed under %smearing. CAUTION: AUTO anchors the occupation to the initial guess for the first five iterations, which is what keeps a SOMO in place on most open-shell systems -- but when the guess ordering is wrong it locks the SCF into the wrong state, and the result then depends on %level_shift and on %initial_guess. O2 (triplet, HF/6-31G(d,p)) is the standard example: AUTO gives -148.57 to -149.45 Ha depending on those settings, STRICT gives -149.611927 Ha, matching ORCA to 1e-9. If an open-shell energy looks too high or moves with %level_shift, re-run with STRICT and compare; %stability check settles it.
%eri_memory_limit_mb	<positive integer, MB>	%eri_mem_limit_mb, %eri_memory_limit, %eri_mem_limit	ERI-buffer memory cap (v0.76.3); falls back to direct JK when exceeded. Default: 4096.
%scf_direct	on | [off]		Force the direct SCF: skip the in-memory ERI store and rebuild integrals every iteration (memory-bound regime of large molecules) (v0.90.2). Default: off.
%scf_incremental	[on] | off		Incremental (delta-P) Fock builds in the direct-SCF path with Schwarz x max|dP| screening; full rebuild every 20 iterations (v0.90.2). Default: on (in direct mode).

Keyword	Options	Synonyms	Designation
%scf_multigrid	off | on | <preset 1..10>	%multigrid	Multigrid SCF (trial mode, v0.99.0): converge on a coarse DFT grid first, then finish on the target %grid from that density. Final energy is converged on the target grid. %job energy only; ignored for HF and CDFT. Default: off.

SCF checkpoint / warm restart (v1.08.0+)

Keyword	Options	Synonyms	Designation
%chkfile	<filename>	%chk_file	Checkpoint file name (case preserved). Default: input file name with the extension replaced by ".chk". The checkpoint stores density matrices (RHF: half density; UHF: Pa, Pb), spin treatment, nbf, natom, charge, multiplicity, element list, geometry (bohr) and the converged energy in a plain-text, portable format.
%chk_write	[on] | off		Write the checkpoint after every successful molecular SCF (each converged optimizer step overwrites it, so after a run it holds the LAST geometry and density). Default: on.
%chk_every	<integer >= 0>		Also write the checkpoint every n iterations DURING the SCF (crash / power-loss recovery for long metallic-cluster runs) (v1.08.3). Default: 0 (converged-only).
			Restart recipe: "%geo read" restores the geometry and "%initial_guess read" restores the density from the checkpoint. Validation on read: hard fail on nbf / natom / element mismatch; warn on charge, multiplicity, geometry.

SCF stability analysis (v1.24.0+)

Keyword	Options	Synonyms	Designation
%stability	check | follow | [off]		SCF stability analysis for real closed-shell (RHF/RKS) references (v1.24.0). Default: off. A converged SCF is a STATIONARY point of the energy, not necessarily a MINIMUM: DIIS reaches saddle points, and a fixed %level_shift or a warm restart can steer it there reproducibly. "check" diagnoses; "follow" also descends. The second variation under orbital

Keyword	Options	Synonyms	Designation
			rotations splits into four channels whose Hessians are the TDDFT (A+B) / (A-B) matrices: real spin-preserving (a lower RHF/RKS solution exists), real spin-breaking (a lower UHF/UKS solution exists), and the two complex channels. All four are reported. A frequent field symptom of an unstable reference is a %tddft run with NEGATIVE excitation energies -- the response problem finding the descent direction the SCF missed. Same scope gates as %tddft: no meta-GGA kernels, no %smearing, no %cdft, no unrestricted references; PCM is supported. Cost is that of a %tddft run on the same system.
%stability_maxcycle	<integer 1..100>	%stability_max_cycle	Maximum number of rotate-and-re-converge cycles under %stability follow. Default: 10. Only the real spin-preserving ("internal") channel is followed: the orbitals are rotated along the most negative eigenvector, C_new = C*exp(kappa), the closed-shell density is rebuilt and the SCF re-converged, until that channel is positive definite or the budget is spent. The external (triplet) channel is reported but NOT followed -- descending it means switching the whole run to an unrestricted reference, which is a user decision (%spin unrestricted, usually with a multiplicity scan).
%stability_step	<real in (0, 1.57] rad>		Rotation step t used in kappa = t*X for the follow descent. Default: 0.35 rad. Along an exact descent direction the energy must go down; a rise is detected and reported, and the usual cause is a fixed %level_shift pulling the SCF back into the original basin (a warning fires before the first cycle in that case). Larger steps widen the escape

Keyword	Options	Synonyms	Designation
			reach from a narrow attractor but do not guarantee escape.

Fractional-occupation smearing (finite electronic temperature, v1.07.0+; gradients/opt v1.09.0, freq v1.09.1)

Keyword	Options	Synonyms	Designation
%smearing	fermi | gauss | mp | cold | [off]	%smear; fermi: fd | gauss: gaussian | mp: mp1, methfessel-paxton | cold: mv, marzari-vanderbilt	Occupation-smearing scheme for the Mermin (finite-T) SCF; HF and DFT, RHF/RKS and UHF/UKS. Default: off. The SCF converges on and reports the Mermin free energy F = E - T*S; the final block prints the bare E, -T*S, the sigma->0 estimate E0 ~ (E+F)/2, the Fermi level(s) and the fractional occupations. Jobs: energy / gradient / opt (v1.09.0; analytical smeared gradients via the fractional energy-weighted density -- Weinert-Davenport -- for RHF/RKS and both UKS moment conventions; %grad_method analytical required) and freq / opt+freq (v1.09.1; semi-numerical route only: FD of the smeared analytical gradient captures the occupation response numerically; %hess_method numerical is auto-selected with a NOTE, an explicit analytical request errors). Frequencies are curvatures of the F(sigma) surface. ts / irc stay gated. No MP2, %tddft or %cdft. Level shift and MOM are bypassed; SAD atomic SCFs run unsmeared; PCM rides along.
%smearing_sigma	<positive real, Ha>	%smear_sigma, %smearing_width	Smearing width sigma in Hartree (for Fermi–Dirac sigma = kB*T_el). Default: 0.01.
%smearing_moment	[fixed] | relaxed	fixed: nupdown | relaxed: relax, shared, common, auto, free	UKS moment convention under smearing (v1.07.1). fixed: two chemical potentials conserve N_alpha and N_beta separately (multiplicity preserved; VASP NUPDOWN analogue). relaxed: one shared mu conserves only N_total; the magnetization relaxes to the free-energy optimum (the input multiplicity

Keyword	Options	Synonyms	Designation
			only seeds the guess). No effect for RKS/RHF.

Density fitting (RI, v0.96.0+)

Keyword	Options	Synonyms	Designation
%ri	[off] | j | jk	j: on, rij | jk: rijk, k	Resolution-of-the-identity (density-fitting) Coulomb / exchange builds. "j": fitted Coulomb for PURE functionals (SVWN, BLYP, BP86, BPW91, PBE, r2SCAN), RKS and UKS; the four-index ERI phase is skipped entirely. "jk" (v0.98.0): fitted J and K, opens HF and global hybrids. Jobs: energy / gradient / opt / freq / opt+freq (analytical RI gradients, v0.97–0.98.3); the analytic Hessian covers pure closed-shell functionals with %ri j (v1.02.0), other cases use the semi-numerical route. Not with range-separated functionals, MP2, %tddft, %cdft, scan / ts / irc / bsse. Auxiliary bases loaded automatically: def2-universal-JFIT (H–Ar; extended, decontracted K–Rn in v1.08.3) for "j", def2-universal-JKFIT (H–Ar) for "jk". PCM rides along. Default: off.
%ri_check	on | [off]		One-shot fitted-vs-exact J diagnostic on the first SCF iteration (prints max|dJ|, dE_J). Validation instrument; costs about as much as a full conventional 2e phase — keep off in production runs.

DFT options

Keyword	Options	Synonyms	Designation
%grid		%grid_quality	DFT integration-grid quality. Default: 3 (standard). No effect for HF or MP2. Presets 1-5 use the Mura-Knowles Log3 radial map; since v1.28.0 its scale is the alpha tabulated by those authors (5.0 Bohr, 7.0 Bohr for groups I and II -- Table IV of J. Chem. Phys. 104, 9848 (1996)) and not the Bragg-Slater radius, which is the scale of the Euler-Maclaurin map and had been substituted by mistake. The correction removed a 2-4 kcal/mol quadrature error on ordinary molecules and is largest for hydrogen-rich ones; see %grid_legacy_mk. Presets 6-10 are

Keyword	Options	Synonyms	Designation
			unchanged and remain bit-identical to earlier releases.
	1		Coarse (25 radial, Mura–Knowles, Lebedev up to 110, no pruning)
	2		Medium (50 radial, 3-tier pruning, up to 302)
	3		Standard (75 radial, 3-tier pruning, up to 590; default)
	4		Extended (100 radial, up to 590)
	5		Extreme (150 radial, up to 590)
	fine	gfine	Preset 6: Gaussian Fine / NWChem (75 radial Euler–Maclaurin, NWChem 5-zone pruning up to 302, Becke size adjustment)
	ultrafine	gultrafine	Preset 7: Gaussian UltraFine (99 radial, NWChem 5-zone pruning up to 590)
	tm	treutler	Preset 8: Treutler–Ahlrichs M4 (75, 302) uniform 302-point angular, no pruning (PySCF-like)
	g09	g09ref	Preset 9: G09 reference (75, 302) uniform 302-point angular, Euler–Maclaurin + Becke size adjustment
	orca7		Preset 10: ORCA Grid-7 reproduction (Treutler–Ahlrichs M3 radial, element-dependent radial count via Krack–Koester, 5-zone pruning {110, 434, 590, 770, 590} with genuine Lebedev-770; v0.78.2/0.81.6)
	response_light (11)		Preset 11: CPKS/TDDFT response grid, light (35 radial Euler-Maclaurin, 3-tier pruning up to 302, Becke size adjustment). Selectable as a production grid, but intended for %cpks_grid.
	response_tight (12)		Preset 12: CPKS/TDDFT response grid, tight (50 radial, otherwise as preset 11).
%xc_deriv	[analytical] | numerical	analytical: anal, exact | numerical: num, fd	Analytical or finite-difference XC functional derivatives during SCF and DFT gradient / Hessian. Default: analytical.
%cpks_grid	same | [light] | tight | <integer 0..12>	%response_grid	Grid on which the XC kernel of the CPKS A-operator (analytical Hessian, polarizability) is integrated (v1.28.0). Default: light = preset 11. "same" reuses

Keyword	Options	Synonyms	Designation
			the production %grid -- always correct, most expensive, and the way to validate a cheaper choice. Up to v1.27 this cache was built by keeping every stride-th point of the production grid; a subset of a Lebedev shell is not a quadrature, the A-operator lost rotational invariance, and (H2O)2 grew imaginary frequencies down to -153 cm^-1. The response grid is genuinely coarser but genuinely valid: full Lebedev shells, correct weights, correct Becke partitioning. Measured on (H2O)2 / 6-311++G(2d,2p) / %grid tm, light (34306 pts) reproduces same (135605 pts) to 0.03 cm^-1 on every mode.
%cpks_mem	<size in GiB; 0 disables>	%cpks_memory	Memory budget for the CPKS XC cache and its per-thread workspaces (v1.28.3). Default: 2.0 GiB. The footprint is estimated before allocating and printed on the "XC cache" line. Over budget, a DEFAULT response grid steps down to the cheapest preset that fits; an EXPLICIT %cpks_grid is honoured with a warning, because silently substituting a quadrature under an explicit request is how the pre-v1.28 defect stayed invisible. The blocked A-product of v1.28.4 removed the nthreads*npoints*nbf term from the workspace, so the remaining cost is the shared cache itself.
%grid_legacy_mk	on | [off]		Restore the pre-v1.28.0 radial scale of the Mura-Knowles presets 1-5 (v1.28.0). Those presets used the Bragg-Slater radius in the Log3 map, which is the scale of the Euler-Maclaurin map, not of Log3: the grid then stopped at 4.4 Bohr around O and 2.6 Bohr around H, and preset 3 -- the default -- integrated N_e to 1.7e-2 and the energy to 3.5 mHa on one water molecule. Provided only so numbers stored before v1.28.0 can be reproduced; not recommended for new work.

Dispersion

Keyword	Options	Synonyms	Designation
%dispersion	NONE | D3	%disp; D3: D3BJ, BJ, D3(BJ)	Empirical dispersion correction. D3 = native D3(BJ) implementation (v1.13.0): energies, gradients and FD-of-gradient Hessians for HF and the common GGA/hybrid functionals. The Chai-Head-Gordon CHG tail of wB97X-D is likewise a native implementation covering H-Xe (v1.13.1-1.13.2) and is enabled automatically with that functional.

MP2 options

Keyword	Options	Synonyms	Designation
%frozen_core	-1 | 0 | <positive integer>	%nfrozen	Frozen-core control. -1 (default) = auto-detect noble-gas core of every non-ECP atom; 0 = correlate all; N = freeze lowest N occupied orbitals (per spin for UMP2).
%pump2	[yes] | no	on, true, 1, T, <bare> | off, false, 0, F	Spin-projected open-shell MP2 (Schlegel-1986 single-annihilation). Default: yes — PUMP2 single-point diagnostic always printed when UMP2 is active. With %pump2 yes, FD-PUMP2 optimization is also enabled (gradient w.r.t. PUMP2 energy). With %pump2 no, PUMP2 is fully suppressed.

Geometry optimization

Keyword	Options	Synonyms	Designation
%geom_maxiter	<positive integer>	%opt_maxiter, %opt_max_iter, %geom_max_iter	Max optimization steps. Default: 50.
%grad_tol	<positive real>		Optimizer convergence: threshold on the max gradient component (Ha/Bohr); the other three criteria scale with it (rms_g = 2/3x, max_dx = 4x, rms_dx = 8/3x; the TS search keeps its looser gradient bases scaled by the same factor). FUNCTIONAL since v1.09.5 -- previously the keyword was parsed but the optimizer workers tested hard-wired thresholds. Default: 4.5e-4, which reproduces the historical criteria set exactly.
%opt_tight	on | [off]	%tightopt	Preset of tighter optimizer convergence thresholds (v1.29.5), equivalent to Opt=Tight in Gaussian. Sets %grad_tol to 1.5e-5, i.e. the four criteria become

Keyword	Options	Synonyms	Designation
			1.5e-5 / 1.0e-5 / 6.0e-5 / 4.0e-5 -- exactly 30x tighter than the defaults, and exactly Gaussian's Tight set, because StudDFT's defaults are Gaussian's and %grad_tol scales all four in the same ratios. (ORCA's !TightOpt tightens only 3x, but its default is already 3x tighter than Gaussian's, so from a common starting point the two agree to within a factor of ~3.) USE IT FOR FREQUENCY JOBS on molecules with soft modes or exploitable symmetry: at the default threshold the optimizer stops with a residual gradient that leaves every soft coordinate displaced by about g/k. On C2H6 (HF/cc-pVDZ) that was enough to break D3d down to C2h -- one H-C-C angle off by 0.047 deg, degenerate modes split by up to 1.3 cm^-1 -- while %opt_tight recovers D3d. Expect a few more cycles. Later %grad_tol overrides it.
%opt_verytight		%verytightopt	As %opt_tight but 300x tighter than the defaults (%grad_tol 2.0e-6), matching the forces of Opt=VeryTight in Gaussian (v1.29.5). Only worth requesting when the gradient itself is clean to that level -- a tight SCF and a good grid -- otherwise the optimizer chases quadrature noise and simply runs out of cycles.
%stop_onerr	(flag)	%stop_on_error, %strict	Strict mode (v1.09.5): abort the run with a nonzero exit code whenever a production SCF fails to converge (single points, optimizer steps, FD-Hessian displacements, scans, IRC) or a geometry/TS optimization exhausts %opt_maxiter. Default (absent): warn and continue, as in earlier releases. SAD/Hirshfeld atomic sub-SCFs and the meta-GGA PBE warm-up are exempt by design.
%grad_method	[analytical] | semianalytical	analytical: full, 1 | semianalytical: semi, fd, 0	Gradient method. PUMP2 gradient is always numerical (h = 1e-3 Bohr).
%opt_coord	[internal] | cart	%opt_coordinates, %opt_coords |	Optimizer coordinate system (v0.75.0). Default: internal (redundant-internals

Keyword	Options	Synonyms	Designation
		internal: internals, redundant, rfo, int | cart: cartesian, bfgs, cart_bfgs, cartesian_bfgs, xyz	RFO). Cartesian path does NOT support constraints or PCM. Diatomics always use Cartesian BFGS.
%initial_hess	[diag] | full	%initial_hessian, %init_hess, %model_hess, %guess_hess; diag: diagonal, lindh | full: almlof, almloef, coupled	Initial model Hessian for the RFO/internal-coordinate minimizer (v0.81.0). diag (default): Lindh diagonal model. full: coupled Almloef/Lindh model with off-diagonal force constants (helps molecules with strongly coupled modes).

Geometry constraints (partial-constraint optimization)

Keyword	Options	Synonyms	Designation
%constrain			Start a constraint block (v0.70+). One declaration per line, terminated by *end or empty line. Compatible with %job opt / opt+freq, internal-coordinate optimizer only.
	B <i> <j>		Freeze bond between atoms i and j
	A <i> <j> <k>		Freeze angle i-j-k (vertex at j)
	D <i> <j> <k> <l>		Freeze dihedral i-j-k-l

PES scan (rigid / relaxed)

Keyword	Options	Synonyms	Designation
%scan			Start a sweep-variable block (v0.69+). Required by %job scan and %job scanopt. One sweep variable per line, terminated by *end or empty line. Requires a %gz Z-matrix geometry.
	<kind> <row> [from] <val_from> [to] <val_to> [step] <val_step>		<kind> = R (bond) | A (angle) | D (dihedral); <row> = 1-based Z-matrix row index. Keywords from/to/step are optional.

Hessian / Frequencies

Keyword	Options	Synonyms	Designation
%hess_method	[analytical] | numerical	analytical: full, 1 | numerical: num, fd, 0	Hessian computation method. Default: analytical (HF, DFT incl. ECP; closed-shell range-separated hybrids since v0.88.2; %ri j with pure closed-shell functionals

Keyword	Options	Synonyms	Designation
			since v1.02.0; static polarizability piggybacks, see %polar). Numerical = semi-numerical finite difference on the analytical gradient (h = 5e-3 Bohr); it is auto-selected (with a NOTE) for MP2/UMP2, PCM, open-shell RS functionals, %smearing, and the remaining RI cases. The numerical route displaces only symmetry-UNIQUE atoms and reconstructs the rest from the point group (v1.09.2; see %hess_nosym).
%hess_nosym	(flag)	%hess_no_sym, %hess_nosymmetry	Disable the symmetry-blocked numerical Hessian (v1.09.2): displace ALL atoms (2 gradients per coordinate) as in earlier releases. By default only symmetry-inequivalent atoms are displaced and the remaining Hessian columns are reconstructed exactly from the point-group operations, H(gC, gA) = R H(C,A) R^T -- CH4 (Td) computes 12 of 30 displaced gradients (2.4x measured). Axis-aligned molecules reproduce the full loop to 1e-12 Ha/bohr^2; arbitrarily oriented DFT cases differ at the Becke-grid rotational-noise level (<= 0.2 cm^-1 at grid 3), with degeneracies preserved BETTER by the symmetric route. When the group is C1 or every atom is unique the loop is bit-identical to the full route.
%hess_step	<positive real, Bohr>	%hess_fd_step	Displacement of the gradient-FD Hessian (v1.28.2). Default: 1.0e-3 Bohr. Affects %hess_method numerical only. It used to be a hard-coded parameter, which made the one convergence check that matters -- does the FD Hessian still move when h changes? -- impossible without a rebuild. Truncation error falls as h^2 while propagated gradient noise grows as 1/h, and on this code the gradient noise is dominated by the moving DFT quadrature, so on soft intermolecular modes the FD Hessian at the default step can be the LESS accurate of the two routes. Vary this before treating a numerical Hessian as the reference: on (H2O)2 / B3LYP / 6-311G(d,p) going 1e-3 -

Keyword	Options	Synonyms	Designation
			> 3e-3 moves the lowest mode by 0.04 cm^-1, i.e. it is converged.

TS options (transition-state search)

Keyword	Options	Synonyms	Designation
%read_hess	<filename>	%read_hessian	.hess file to load before TS search (v0.72.0) — and REQUIRED by %job irc (v0.86.0). Reader verifies geometry, method, basis, ECP, charge, multiplicity, spin, format version (for IRC the file geometry is authoritative and coordinates are deliberately not compared). If omitted for a TS search, the TS driver computes the Hessian itself at the input geometry.
%read_geohess	<filename>	%read_geomhess, %read_hess_geom	Same as %read_hess, plus: the GEOMETRY is taken from the same .hess file (v1.30.0). It sets %read_hess as well, so the TS optimizer, the IRC driver and the frequency driver need no separate keyword. The substitution happens before the molecule is built, so the input echo, the point-group detection, the basis/ECP assignment and every downstream stage all see the file geometry. With this keyword the %geo / %gz block becomes OPTIONAL - which is the point for an IRC restart: the TS coordinates already live in the .hess file, and copying them by hand is a chance to shift the starting point off the saddle. If a geometry IS given, the atom count and the atomic numbers must match the file element by element (a permuted atom list would pair Hessian rows with the wrong nuclei) and only the coordinates are replaced; the largest shift is reported, so a stale pasted geometry is visible in the log. Any inconsistency is fatal - continuing with the input geometry after the user asked for the file geometry is exactly the silent error the keyword exists to prevent.
%ts_mode	<positive integer>	%tsmode	1-based mode index to follow on the first P-RFO step (v0.72.0). After step 1,

Keyword	Options	Synonyms	Designation
			optimizer switches to maximum-overlap mode tracking. Default: 1 (lowest mode).
%ts_recalc_hess	<non-negative integer>; [0]	%ts_recalc, %recalc_hess	Recompute the EXACT Hessian every N steps of a P-RFO transition-state search instead of Bofill-updating it (v1.30.1); the Bofill update still runs on the intermediate steps. Default: 0 = never, for every backend - one recompute costs a CPKS solve or 6*natom gradients, so making it automatic would slow down the majority of searches that converge on updates alone. The backend follows %hess_method (analytical CPKS, otherwise gradient FD), so the recomputed matrix is built exactly like the initial one. WHEN TO USE IT: when a search misbehaves, and the symptom is specific - max|g| passes through a minimum and grows again, usually reported together with a negative-eigenvalue count other than one. Bofill is a rank-two update: it reproduces the curvature along the last step and leaves every other direction untouched, so over a long walk the stored matrix drifts, and near a saddle the drift changes the number of negative eigenvalues. P-RFO maximizes along one eigenvector and minimizes along the rest, so once the matrix carries three negative eigenvalues the search is maximizing along the wrong direction and walks away from a saddle it had already reached. Measured on H2CO -> H2 + CO (BLYP/cc-pVDZ): max|g| fell to 1.6E-03 at step 16, then grew back to 3.4E-02 by step 27. N = 5 is a reasonable starting value; N = 1 is exact Newton at every step and is only worth it on small systems. The interim recomputes DO print their frequency table on purpose - the mode count at that moment is how one sees whether the search still tracks the right mode - but they do NOT write .hess / .molden files, so the one written at the located TS is not buried under numbered siblings. If the backend

Keyword	Options	Synonyms	Designation
			refuses, the search keeps the Bofill update and says so.

IRC options (reaction-path following, v0.86.0+)

<table>
<tr><th>Keyword</th><th>Options</th><th>Synonyms</th><th>Designation</th></tr>
<tr><td>%irc_step</td><td><real in (0, 2]></td><td></td><td>Path step size in amu^1/2*bohr (GS2 pivot at s/2, corrector sphere radius s/2). Default: 0.10.</td></tr>
<tr><td>%irc_maxpoints</td><td><integer in [1, 10000]></td><td>%irc_max_points</td><td>Max path points per branch. Default: 25.</td></tr>
<tr><td>%irc_gtol</td><td><positive real></td><td></td><td>Termination threshold on max |dE/dx| (Ha/bohr): "minimum region reached". Default: 4.5e-4 (same scale as %grad_tol).</td></tr>
<tr><td>%irc_direction</td><td>forward | reverse | [both]</td><td>%irc_dir; forward: f | reverse: backward, r, b</td><td>Which branch(es) of the path to follow from the TS. Default: both.</td></tr>
<tr><td>%irc_mode</td><td><positive integer></td><td></td><td>Which imaginary mode to start along (1-based, by ascending Hessian eigenvalue). Default: 1 (correct for every true TS).</td></tr>
<tr><td>%read_geohess</td><td><filename></td><td>%read_geomhess, %read_hess_geom</td><td>See the TS options table. A .hess file written at a converged TS already stores the saddle geometry, so "%read_geohess ts.hess" starts an IRC with NO geometry block at all (v1.30.0). Recommended over %read_hess + a pasted %geo block: an IRC started a little off the saddle is the failure mode this replaces. Note that %job irc requires a Hessian file either way, and that the gradient at path point 0 is now computed, printed and graded against %irc_gtol (v1.30.0) - a branch started from a STALLED TS search is reported instead of quietly retracing its partner.</td></tr>
</table>

TDDFT / TDHF excited states (v0.87.0+)

Keyword	Options	Synonyms	Designation
%tddft	<nstates, 1..1000>		Compute <nstates> excited states by linear-response TDDFT / TDHF (Casida) after the SCF. Requires %job energy and a closed-shell (RHF/RKS) reference; not with MP2, %cdft, %smearing or meta-GGA (r2SCAN) kernels. PCM is supported via the LR-PCM response kernel (v0.94.0).

Keyword	Options	Synonyms	Designation
			Output: state table (Ha / eV / nm / oscillator strength) + dominant amplitudes + RPA de-excitation diagnostics. Since v1.15.0-1.16.0 the singlet LDA/GGA/hybrid kernel uses the rank-1 amplitude fast path and all MO half-transforms are dgemm-major: on transition-metal systems the TDDFT step is 30-80x faster than in 1.09.6-1.15.x (no input changes needed).
%tddft_tda	on | [off]		Tamm–Dancoff approximation (CIS-like) instead of full RPA-type TDDFT. Default: off.
%tddft_maxdim	<integer; 0 = automatic>	%tddft_max_dim	Hard cap on the Casida/TDA subspace dimension (v1.16.2). Default: 0, i.e. the automatic byte-based forecast picks the dimension from the available memory. A positive value overrides the forecast; use it when the automatic estimate is too optimistic for the machine, or to reproduce a stored spectrum on different hardware.
%tddft_multiplicity	[singlet] | triplet | both	%tddft_mult; s, t	Spin symmetry of the computed excitations. Default: singlet. Triplet oscillator strengths are 0 (spin-forbidden).
%tddft_dipmom	on | [off]		Transition dipoles <0|mu|n> and <m|mu|n>, unrelaxed excited-state dipole moments, and a two-state Generalized Mulliken–Hush electron-transfer coupling table (v0.87.1). Default: off (cheap).
%export_tddft	on | [off]	%tddft_export	Write <base>_tddft_mo.molden (ground-state MOs) and <base>_tddft_states.txt (Gaussian-layout excitation blocks with CI amplitudes) for MultiWfn / THEODORE hole–electron / NTO analyses (v0.87.1). Default: off.
%tddft_pcm	nonequilibrium | equilibrium	noneq, neq | eq	Solvation regime of the LR-PCM TDDFT kernel (v0.94.0). nonequilibrium: kernel at the optical permittivity eps_inf = n^2 (vertical

Keyword	Options	Synonyms	Designation
			absorption). equilibrium: kernel at the static eps (full solvent relaxation).
%eps_optical	<real >= 1>		Optical permittivity eps_inf for the nonequilibrium LR-PCM kernel. Resolution: explicit %eps_optical > built-in n^2 table (18 named solvents) > 2.0 fallback with a NOTE.
%tddft_selftest	on | [off]		Runtime numerical self-tests of the kernel contraction machinery, incl. the (A+B)-column vs CPKS identity and the LR-PCM response check (teaching / debugging aid). Default: off.

PCM (implicit solvent, v0.73.0+)

Keyword	Options	Synonyms	Designation
%pcm	none | cpcm | iefpcm | cosmo	cpcm: c-pcm, c_pcm | iefpcm: ief-pcm, ief_pcm, pcm | none: off, no, false, 0, F	Continuum solvation model. Default: none. %solvent or %eps auto-activate IEFPCM if no %pcm is given (v0.74.0; matches Gaussian SCRF=PCM — note this changed from the earlier C-PCM auto-default). Supported with PCM: %job energy, gradient, opt (analytical PCM gradients, v0.74.1), freq, opt+freq (semi-numerical PCM Hessian, v0.93.0), and %tddft (LR-PCM, v0.94.0). RKS and UKS; all functional classes incl. range-separated (v0.92.0). Not with MP2 or %job bsse / scan / ts / irc.
%solvent	<name>		Named solvent (resolved to tabulated permittivity; overrideable by %eps). Recognized: water (h2o), methanol (meoh), ethanol (etoh), acetone, acetonitrile (mecn), dmso, dmf, thf, chcl3 (chloroform), ccl4, dichloromethane (dcm, ch2cl2), toluene, benzene, cyclohexane, hexane (n-hexane), diethylether

Keyword	Options	Synonyms	Designation
			(et2o, ether), ethyl_acetate (etoac), ammonia (nh3), vacuum (none).
%eps	<real >= 1.0>	%epsilon, %pcm_eps	Direct override of dielectric permittivity. Vacuum = 1.0.
%pcm_lebedev	<positive integer>		Per-atom Lebedev quadrature order on cavity surface. Default: 110. Allowed: 6, 14, 26, 38, 50, 74, 86, 110, 146, 170, 194, 230, 266, 302, 350, 434, 590, 770.
%pcm_scaling	<positive real>		Scaling factor on Bondi vdW radii for atomic-sphere cavity. Default: 1.20 (standard COSMO/CPCM).
%pcm_switch	<positive real, Bohr>		Width of cubic-taper switching function on cavity-tessera occupancy. Default: 0.20 Bohr (typical 0.1 - 0.4).
%pcm_grad_method	[analytical] | numerical	%pcm_gradient_method | analytical: ana, an, 1, true, yes, on | numerical: num, fd, 0, false, no, off	PCM gradient method (v0.74.x). Default: analytical (fast). Numerical = central-difference FD (6N+1 PCM SCFs), ~10-30x slower; agrees with analytical to ~1e-5 a.u./bohr; retained for cross-validation.

Properties and analysis

Keyword	Options	Synonyms	Designation
%pop_analysis	[on] | off	%population, %pop	Extended population analysis after the always-on Mulliken charges (v1.05.0): Löwdin charges; Hirshfeld charges (grid integration against the program's own spherically averaged SAD promolecule) + CM5 charges (parametrized for Z <= 20, warning above); Mayer / Mulliken (overlap) / Löwdin (Wiberg) bond orders; Mayer bonded / free / total valences; Mulliken, Löwdin and Hirshfeld spin populations for UHF. Bare keyword = on; default (absent): off.
%symmetry	[on] | off	%sym, %pointgroup, %point_group	Point-group symmetry analysis (v1.06.0). NOTE (v1.09.2): the point group itself is now detected and reported on EVERY run (one line, no keyword), re-detected after every optimization with a change notice, and drives

Keyword	Options	Synonyms	Designation
			the symmetry-blocked numerical Hessian. This keyword additionally enables the detailed report: detects the molecular point group and labels canonical MOs and vibrational normal modes by irreducible representation; prints the electronic configuration and the Gamma_vib summary. Supported: C1, Cs, Ci, Cn/Cnv/Cnh/Sn/Dn/Dnd/Dnh (n <= 8), Td, Oh, and linear C∞v / D∞h; icosahedral and rarer cubic groups are analyzed in their largest supported axial subgroup with a printed note. The geometry is never reoriented or symmetrized — all energies are unchanged. Default (absent): off.
%symmetry_tol	<positive real, bohr>	%sym_tol, %symtol	Atom-matching distance tolerance for detecting a symmetry operation. Default: 1e-3.
%polar	[on] | off	%polarizability	Static dipole polarizability from a 3-RHS field CPKS solve, computed alongside the analytic frequencies (%job freq / opt+freq on the analytical-Hessian route) (v1.04.0). Closed shell only (UHF/UKS prints a NOTE and skips). Prints the Gaussian "Dipole polarizability, Alpha" block (iso, aniso, lower triangle; au / esu / SI). Works with %ri j and hybrids. Default (absent): off.
%efield	<fx> <fy> <fz> (au)	%field	Uniform external electric field added to H_core before the SCF (finite-field properties; also the FD oracle for %polar) (v1.04.0). The dipole print then reports mu(F). Default: 0 0 0.

Output control

Keyword	Options	Synonyms	Designation
%print_mo	(flag)		Print converged MO coefficient matrix at end of SCF.
%print_1e	NONE or <filename>	%print_1e_integrals	Print 1e matrices (S, T, Vnuc, Hcore) in the WORKING basis of the run (spherical if %spherical, else Cartesian) to stdout, or to <filename> if given (v0.84.1).
%print_time	(flag)	%print_timing	Print detailed timing breakdown per timer counter at end of run.
%print_grad	(flag)	%print_gradient	Print per-atom Cartesian gradient at every optimization step (format: <Element><index> gX gY gZ).

Keyword	Options	Synonyms	Designation
%export_mo	(flag)	%molden_mo, %export_orbitals	Save converged MO coefficient matrix in Molden format to the external file <base>_mo.molden (Cartesian convention, viewer-ready per-component normalization; v0.82.0). Written for every job type that ends with a single wavefunction (energy, gradient, opt, freq, optfreq, ts, ts+freq); for MP2 the canonical HF orbitals are exported.
%export_1e	(flag)	%export_1e_integrals	Save 1e matrices (S, T, Vnuc, Hcore) to <base>_1e.export in the SAME Cartesian basis, AO order and normalization as the %export_mo molden file, so C_A^{T} M C_B with molden coefficients is convention-consistent (intended for CDFT S_AB / H_AB coupling construction; v0.84.1).

Comments

Keyword	Options	Synonyms	Designation
# or !			"#" at the start of a line: whole line is ignored. "!" anywhere on a line: the rest of the line is ignored — in-line comments ARE supported since v0.74.0 (e.g. "%job opt ! comment"). TAB, CR and other non-printable characters are replaced by spaces before parsing.

Part II. Full Edition only

In the Student Edition the keywords of this part stop the run with the message quoted above.

Constrained DFT (v0.84.0+; Full Edition)

Keyword	Options	Synonyms	Designation
%cdft	[becke]		Start a charge-constrained DFT block (Wu–Van Voorhis; Becke atomic weights are the only scheme in this release). One fragment definition per line; the block ends at a blank line, *end, or the next %-keyword. %job energy only (gradients / optimization are a later stage); not with %tddft, %smearing, %ri or range-separated functionals. The reported energy is the physical E_KS at the constrained density; converged Lagrange multipliers and CDFT orbital energies are printed.

Keyword	Options	Synonyms	Designation
	<label> <atom indices...> charge <q_target>		Fragment definition line, e.g. "Donor 1 2 3 charge +1.0". Label is free text; atom indices are 1-based; the literal keyword "charge" separates the atom list from the target fragment charge. Multiple independent fragments are handled through the full Newton system (do not let two fragments cover ALL atoms — the Jacobian becomes singular).
%cdft_tol	<positive real>		Outer-loop tolerance on max |N_c - N_target| (electrons). Default: 1e-4.
%cdft_maxiter	<positive integer>		Max outer (Lagrange-multiplier) Newton iterations. Default: 50.
%cdft_spin	[auto] | restricted | unrestricted	r, rks, rhf | u, uks, uhf	Spin treatment of the CDFT diabat (v0.84.1). auto: RKS for a closed-shell singlet (robust), UKS otherwise. restricted forces RKS (error if open-shell); unrestricted forces UKS (e.g. broken-symmetry biradical diabats; the historical v0.84.0 behaviour).
%cdft_level_shift	<non-negative real, Ha>	%cdft_levshift	Floor for the Roothaan level shift of the inner constrained SCF (pure stabiliser: converged density / energy / Jacobian unchanged). Effective shift = max(%level_shift, %cdft_level_shift). Default: 0.2 (RKS) / 0.5 (UKS); crank up (1.5–2.0) for hard open-shell diabats; 0 disables.
%cdft_initial_guess	read | [off]	read: chk, restart | off: none	Warm-start policy of the constrained SCF: READ starts from the checkpoint file (%chkfile) of a previous (typically unconstrained) run; OFF uses the standard %initial_guess route.

Real-time TDDFT (v1.10.0+; Full Edition)

Keyword	Options	Synonyms	Designation
%rt_tddft	<nsteps>	%rt, %rttddft	Enable real-time TDDFT: propagate the converged SCF (or CDFT-released) state for <nsteps> electronic steps (v1.10.0+). Works for RKS and UKS, with %smearing (fractional occupations) and after a %cdft preparation (the CDFT-release charge-transfer protocol). Writes <base>_rt.dat (energy, dipole, idempotency and

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			reference-overlap monitors) and the population/spectrum files below.
%rt_dt	<dt, a.u.>	%rt_timestep	Electronic time step in atomic units.
%rt_prop	[midpoint] | magnus2 | etrs | mmut | em	%rt_propagator	Propagator: exponential midpoint with predictor-corrector (default), 2nd-order Magnus, ETRS, modified-midpoint MMUT, or plain exponential.
%rt_pc_maxiter	<positive integer>		Predictor-corrector iteration cap per step.
%rt_pc_thresh	<positive real>		PC convergence threshold on max|dP| between corrector passes.
%rt_print_every	<positive integer>		Console monitor stride.
%rt_init	[ground] | homo_lumo		Initial electronic state: the SCF ground state or a HOMO->LUMO-promoted determinant.
%rt_kick	<kappa, a.u.>		Delta-kick amplitude applied at t = 0 (absorption-spectrum protocol).
%rt_kick_dir	x | y | z | [xyz]		Kick direction; XYZ runs the three Cartesian kicks and averages the spectrum.
%rt_field	<fx> <fy> <fz>		Uniform external electric field (a.u.) acting during the propagation.
%rt_field_omega	<w, Ha>		0 = static field; w > 0 turns %rt_field into the amplitude of a harmonic drive E(t) = F sin(wt) (resonant-Rabi setups).
%rt_field_tmax	<t, a.u.>		Switch the field off for t > tmax (pump-then-watch runs) (v1.11.11).
%rt_ramp	<tau, a.u.>		sin^2 switch-on ramp of the field over tau.
%rt_gamma	<g, a.u.>		Lorentzian damping applied in the dipole Fourier transform.
%rt_wmin	<w, Ha>		Lower edge of the spectral window.
%rt_wmax	<w, Ha>		Upper edge of the spectral window.
%rt_dw	<dw, Ha>		Frequency-grid step of the printed spectra.
%rt_nmodes	<integer, 1..15>		Number of charge-beat modes in the per-atom charge-mode decomposition table (density normal modes; the full per-column spectra go to <base>_rt_pop_spectrum.dat).
%rt_print_pop	<atom list>		Atom selection for the Mulliken population trajectory <base>_rt_pop.dat; default: all atoms when natom <= 15. CDFT Becke

Keyword	Options	Synonyms	Designation
			fragment charges are appended automatically when constraints are defined.
%rt_charge	<q> <x> <y> <z>		External point charge: value in a.u., position in Angstrom. Repeat the keyword for several charges.
%rt_qvel	<iq> <vx> <vy> <vz>		Velocity (a.u.) of external charge iq; a nonzero velocity turns on the moving-charge driver (projectile/scanning protocols) (v1.11.14).

Ehrenfest dynamics (v1.11.0+, smeared v1.12.0; Full Edition)

Keyword	Options	Synonyms	Designation
%ehrenfest	<nsteps>		Mean-field (Ehrenfest) nonadiabatic dynamics: <nsteps> nuclear steps with self-consistent electron-nuclear propagation (v1.11.0+); <nsteps> = 0 runs the FORCE-CHECK oracle mode (single-point force cross-validation). With %smearing the fractional occupations are transported exactly by the spectral scheme of v1.12.0 (smeared Ehrenfest; RKS and UKS).
%eh_dt	<dt, a.u.>		Nuclear time step (default 10 a.u.).
%eh_esub	<positive integer>		Electronic substeps per nuclear step (default 10).
%eh_veloc	<iatom> <vx> <vy> <vz>		Initial velocity (a.u.) of atom iatom; repeat per atom.
%eh_temp	<T, K>		Maxwell-Boltzmann sampling of the initial nuclear velocities at temperature T (v1.11.3).
%eh_seed	<integer>		RNG seed of the thermal sampling (default 20260721; fix it for reproducible trajectories).
%eh_comoving	on | [off]		Comoving (atom-following) Gaussian-basis treatment of the electronic propagation (v1.11.3).
%eh_print_every	<positive integer>		Console print stride.
%eh_dterm	on | [off]		EXPERIMENTAL nonadiabatic D-term coupling; default off.
%eh_fd_force	on | [off]		Extend the force-check oracle with finite-difference forces (validation runs) (v1.11.5).
%eh_skip_wa	on | [off]		Drop the antisymmetric W-term of the equations of motion (diagnostic only).

Getting started (obtaining program, setting up and start calculations)

Windows

Go to the web site of the program (GitHub or other).

1. Open `StudDFT_<version>_Win64.zip` and press the `Download raw file` button on the right.
   Place the archive on a local disk and unpack it. You will get a directory such as `c:\StudDFT` with the programs inside.
   Do not unpack into a protected Windows folder ("Program Files", "Documents"), into a OneDrive-synchronised folder, or into a path containing Cyrillic characters or spaces.

2. Run `set_user_environment.bat` to set the PATH and STUDDFT variables for the current user. You can also set them by hand to the directory where the program resides. Re-login or restart for the change to take effect.

3. If needed, open `sd.bat` in any plain-text editor (Far, Notepad, Notepad++, ...) and adjust the OpenMP settings:

```
set OMP_NUM_THREADS=4       <-- number of physical cores to use
set OMP_STACKSIZE=64M       <-- OpenMP stack size per thread
```

   Use the number of PHYSICAL cores, not logical ones: this code is memory-bandwidth bound in the integral and XC stages, and hyper-threading typically costs more than it gains. Raise OMP_STACKSIZE if a large job stops with a stack overflow.

4. Go to the working directory that holds your input file and run:

```
sd h2o.inp
```

   After a successful run the output file `h2o.log` appears next to the input. Auxiliary files (`h2o.chk`, `h2o.hess`, `h2o.molden`, ...) are written to the same place.

Linux

If a binary package is provided, unpack it, add its directory to PATH and run `studdft h2o.inp`. Otherwise build from source:

1. Prerequisites: a Fortran 2008 compiler (gfortran 9 or newer, or Intel Fortran) and LAPACK/BLAS. On Debian/Ubuntu:

```
sudo apt install gfortran liblapack-dev libblas-dev
```

2. The source tree ships paired files of which exactly ONE of each pair must be compiled: the edition pair (`edition_full.f90` / `edition_student.f90`, and likewise `cdft_`, `rt_tddft_`, `ehrenfest_`) and the platform pair (`sys_memory_posix.f90` / `sys_memory_win.f90`, `sys_hostinfo_posix.f90` / `sys_hostinfo_win.f90`). On Linux take the `_posix` variants.

3. Compile in module-dependency order. Sources must be compiled after every module they `use`; a build system that ignores this fails with a missing `.mod` file, and the error names the CONSUMER of the module, not the file that failed to build. The supplied makefile encodes a correct order:

```
make            # produces ./studdft
./studdft h2o.inp > h2o.log
```

4. Set the OpenMP environment before running:

```
export OMP_NUM_THREADS=4
```

```
export OMP_STACKSIZE=64M
```

On Linux the program writes its report to standard output, so redirect it to a file as shown above; auxiliary files are still created next to the input.

Documentation
See `Short Manual_<version>.pdf` for the keyword list of StudDFT.
See `../docs/` for additional documentation.
See `../examples/` for sample inputs.
Download and unpack `tests` to check the results for 50+ molecules with various methods.

Input files and formats

An input file is plain ASCII. Each line is either a keyword line, a line inside a block, a comment, or blank.

- Keyword lines start with % and carry at most one keyword: "%keyword value". Keywords and their option words are case-insensitive; file names are not.
- Comments start with # or ! and may appear anywhere on a line; the rest of the line is ignored.
- Blank lines are ignored, EXCEPT inside a block, where a blank line closes the block.
- Block keywords are %geo, %gz, %scan, %constrain and %cdft. %geo, %scan, %constrain and %cdft close on a blank line or on a line reading *end; %gz closes on a BLANK LINE ONLY, so always leave an empty line after a Z-matrix. Since v1.30.2 a Z-matrix may also be written inside %geo - it is recognized from a first data line carrying exactly one token - and such an auto-detected block closes on *end as well as on a blank line.
- Coordinates in %geo are in Angstrom by default; write "%geo bohr" for atomic units. A Z-matrix is always in Angstrom and degrees.
- Defaults when omitted: %charge 0, %mult 1, %job energy, %method hf.
- The report is written to <base>.log on Windows (the sd.bat wrapper redirects it) and to standard output on Linux. Auxiliary files use the same base name: <base>.chk, <base>.hess, <base>.molden.

Example input files

Every example below is complete and runnable as it stands. Replace the geometry with your own; keep the blank line that closes each %geo block.

1. Geometry optimization and frequencies for H2O, B3LYP/6-31G(d,p)

```
# Comments marked with # or !
%method  b3lyp            ! Commands start with %
%basis   6-31G(d,p)       ! Comment can be on any place of line
%job     opt+freq
%charge  0
%mult    1
%geo      ! Cartesian coordinates (Anstroems), ends at a blank line:
O   0.000000   0.000000   0.117370
H   0.000000   0.756950  -0.469483
H   0.000000  -0.756950  -0.469483
```

2. Explicit variation of basis sphericity, UHF/UKS model, grid type, and geo units:
```
%method  blyp
%job     energy          ! Single-point calculations
%basis   cc-pVDZ
%spherical               ! Basis of spherical 5d/7f functions
                         ! Default - Cartesian 6d/10f
%spin    unrestricted    ! UHF/UKS model (default if mult>1)
%grid    fine            ! Gaussian Fine grid (default - Standard)
                         ! Blank lines are ignored
#%charge  0    ! default is 0, so no need to indicate it
#%mult    1    ! default is 1, so no need to indicate it

%geo bohr                ! Geometry in atomic units, not Angstroems
O   0.00   0.00   0.20
H   0.00   1.50  -0.92
H   0.00  -1.50  -0.92
```

2. Rigid scan in Z-matrix coordinates
```
%method  mp2
%basis   6-31G(d,p)
%job     scan

%gz                       ! Z-matrix; MUST end with a blank line
H1                        ! Since v.1.30.2 %gz is not necessary
O2 H1  0.95               ! %geo recognizes Z-matrix automatically
H3 O2  0.95  H1  105.

%scan                     ! one scanned variable per line
A 3  from 100. to 110. step 1.
```

3. Warm restart from a checkpoint (two runs)
The first run converges the SCF and writes h2o.chk. Restarting from it saves the whole SCF of the second run, which matters when the second job is expensive (frequencies, TDDFT) or when the first run was hard to converge. "%geo read" takes the geometry from the checkpoint, so the %geo block is not repeated.
Run 1 (input file a.inp): optimize and leave a checkpoint
```
%method     b3lyp
%basis      6-31G(d,p)
%job        opt
%chkfile    h2o.chk           ! default is <input>.chk
%chk_write  on                ! default; written after every SCF

%geo
O   0.000000   0.000000   0.117370
H   0.000000   0.756950  -0.469483
H   0.000000  -0.756950  -0.469483
```

Run 2 (input file b.inp): frequencies at the optimized geometry
```
%method     b3lyp
%basis      6-31G(d,p)
```

```
%job        freq
%chkfile    h2o.chk      ! use checkpoint file from previous run
%geo read                ! geometry from the checkpoint
%initial_guess read      ! density from the checkpoint
```

4. Transition-state search, then a Hessian for it

A TS search converges far more reliably from a computed Hessian than from a model one. The first run produces a.hess; the second reads it and follows the imaginary mode. %ts_mode picks which mode to follow. It is worth setting when the guess structure has more than one soft mode. %ts_recalc_hess is shown for completeness and is NOT needed by a well-behaved search (its default is 0). Add it only when a run misbehaves in the specific way described in the TS options table: max|g| falls, passes through a minimum and then grows again. The value is a step interval, so 5 asks for one exact Hessian every fifth step. "%read_geohess a.hess" reads the proper guessed TS coordinates from a.hess, so nothing has to be copied by hands.

```
! Run 1 (file a.inp): initial Hessian at the guessed TS structure
%method     b3lyp
%basis      6-31G(d,p)
%job        freq

%geo           ! Guessed TS geometry(Cartesian or Z-matrix):
N1
C2 N1  1.2
H3 C2  1.   N1   70.

! Run 2 (file b.inp): TS search using initial Hessian
%method     b3lyp
%basis      6-31G(d,p)
%job        ts+freq           ! search, then freq verification
%read_geohess  a.hess         ! read geo and hess from a.hess
%ts_mode    1
%ts_recalc_hess 5             ! default 0 = never; see above
```

5. IRC from a located TS

The IRC walk REQUIRES a Hessian file calculated in the previous run (file b_ts.hess, see above). Gas-phase HF/DFT only.

```
%method       b3lyp
%basis        6-31G(d,p)
%job          irc
%read_geohess    b_ts.hess    ! read located TS geo and hess
%irc_direction both           ! forward | reverse | both
%irc_maxpoints 40
%irc_step     0.10
```

6. Counterpoise (BSSE) correction

With %job bsse every geometry line carries a fifth column, the fragment index, numbered contiguously from 1. The run reports the complex, the fragments in their own basis and the fragments in the full dimer basis, and the Boys-Bernardi corrected interaction energy.

```
%method  b3lyp
%basis   6-31G(d,p)
%job     bsse
```

```
%dispersion d3bj              ! sensible for a weak complex

%geo                          ! X Y Z <fragment>
O   -1.551  -0.114   0.000   1
H   -1.934   0.762   0.000   1
H   -0.599   0.040   0.000   1
O    1.350   0.111   0.000   2
H    1.680  -0.373   0.754   2
H    1.680  -0.373  -0.754   2
```

5. Implicit solvent (PCM)

Give either a named solvent or an explicit permittivity with %eps. IEFPCM is the rigorous dielectric model; C-PCM and COSMO use the conductor approximation, which is indistinguishable from IEFPCM at water-class permittivity but overestimates the screening by ~16 % at benzene-class values.

```
%method   b3lyp
%basis    6-31G(d,p)
%job      opt
%pcm      iefpcm             ! none | cpcm | iefpcm | cosmo
%solvent  water              ! or: %eps 78.36

%geo
O   0.000000   0.000000   0.117370
H   0.000000   0.756950  -0.469483
H   0.000000  -0.756950  -0.469483
```

NOTE: %pcm is not necessary if you indicate %solvent (IEFPCM model will be used by default).

7. Excited states (TDDFT) with exports

TDA is cheaper and more robust than full RPA; drop %tddft_tda for the Casida problem. %tddft_pcm adds the solvent response of the excitation. The export writes ground-state MOs and the transition densities in Molden format for external viewers.

```
%method       b3lyp
%basis        6-31+G(d)
%job          energy
%tddft        10          ! number of excited states
%tddft_tda    on          ! Tamm-Dancoff approximation
%tddft_dipmom on          ! transition dipoles / oscillator strengths
%pcm          iefpcm
%solvent      water
%tddft_pcm    equilibrium     ! solvent response for the excitations
%export_tddft on

%geo
O   0.000000   0.000000   0.117370
H   0.000000   0.756950  -0.469483
H   0.000000  -0.756950  -0.469483
```

8. Exporting integrals and orbitals

Useful for feeding an external code or for checking the AO basis. The 1e matrices are written in the SAME Cartesian AO order the program uses internally, which is not necessarily the order of another program.

```
%method     hf
%basis      6-31G(d,p)
%job        energy
%export_1e     ! export S, T, Vnuc, Hcore -> file <base>_1e.export
%export_mo     ! export MO coefficients   -> Molden file
%print_mo   on      ! print MO to the log-file
%pop_analysis on    ! population analysis beyond Mulliken:
                    ! Loewdin/Hirschfeld/CM5 charges, Mayer BO etc
%geo
O   0.000000   0.000000   0.117370
H   0.000000   0.756950  -0.469483
H   0.000000  -0.756950  -0.469483
```

9. Open-shell species: a caution

The default occupation policy anchors the orbital occupation to the initial guess for the first five SCF iterations. When the guess ordering is wrong this locks the SCF into the wrong state and the energy then depends on %level_shift and %initial_guess. O2 is the standard example. If an open-shell energy looks suspicious, re-run with STRICT and compare.

```
%method          hf
%basis           6-31G(d,p)
%job             opt+freq
%mult            3
%uks_occupation  strict       ! plain Aufbau, no MOM anchoring

%geo
O   0.0   0.0   0.00
O   0.0   0.0   1.21
```

10. Constrained DFT (Full Edition)

One fragment definition per line: a free-text label, the 1-based atom indices, the literal word "charge", and the target fragment charge. Do not let the fragments cover ALL atoms - the Newton Jacobian becomes singular. %job energy only in this release.

```
%method    b3lyp
%basis     6-31G(d,p)
%job       energy
%charge    0
%mult      1
%cdft      becke
     Donor    1 2 3 4    charge  +1.0
%cdft_tol  1e-5
%cdft_spin auto

%geo
...DA complex geometry...
```

11. Ehrenfest dynamics (Full Edition)
Mean-field nonadiabatic dynamics: nuclei move classically on the evolving electronic density. %ehrenfest 0 runs the force-check oracle (a single-point cross-validation of the forces) and is the right first step before a production trajectory. Fix %eh_seed to make a thermally sampled trajectory reproducible.

```
%method       b3lyp
%basis        6-31G(d)
%ehrenfest    500           ! nuclear steps (0 = force-check oracle)
%eh_dt        10              ! nuclear time step, a.u.
%eh_esub      10              ! electronic substeps per nuclear step
%eh_temp      300            ! Maxwell-Boltzmann initial velocities,
K
%eh_seed      20260721        ! fix for a reproducible trajectory
%eh_print_every 10

%geo
...geometry...
```

12. Checking that an SCF solution is a minimum
A converged SCF is a stationary point, not necessarily a minimum. Run this whenever an open-shell or charge-transfer energy looks too high, when the result moves with %level_shift, or when a TDDFT run returns a negative excitation energy - that is the response problem finding the descent direction the SCF missed.

```
%method             b3lyp
%basis              6-31G(d,p)
%job                energy
%stability          follow    ! check = diagnose only
%stability_maxcycle 10
%stability_step     0.35

%geo
...geometry...
```